

Essence: A Personal Embedding Cluster Framework for Depth-Optimized Recommendations

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Abstract

We present **Essence**, a personal embedding cluster framework for depth-optimized content recommendations. Unlike collaborative filtering (CF), which derives recommendations from population-level interaction patterns, Essence operates exclusively within a single user’s engagement history. Consumed items are embedded using a pre-trained sentence encoder and clustered into K semantic centroids via K -means, each representing a distinct interest profile. Recommendations are generated by selecting the *active centroid* aligned with the user’s recent context and performing cosine-similarity search against unseen items. We evaluate Essence on the Last.fm-1K dataset against a popularity-based CF baseline and a content-based average embedding baseline. Essence achieves Recall@10 of 0.0616 and Long-Tail Recall@10 of 0.0146, outperforming CF (0.0081 / 0.0000) and the average embedding baseline (0.0414 / 0.0032). The $4.5\times$ improvement in long-tail recall empirically validates that active centroid selection surfaces niche items more effectively than averaged user representations, without requiring any cross-user data. Code: <https://github.com/biditdas18/essence>

1 Introduction

Modern recommendation systems are built on a foundational assumption: that a user’s future preferences can be predicted from the preferences of users who behaved similarly in the past. Collaborative filtering (CF) — in its many forms — derives its signal from cross-user interaction patterns. This achieves strong performance on standard benchmarks, particularly on metrics dominated by popular items.

However, this population-level orientation introduces a systematic bias: CF optimises for the median user, surfacing broadly consumed items rather than personally resonant ones. A user who engages deeply with obscure folk music and niche science fiction receives recommendations shaped by thousands of loosely similar users. The result is well-documented: CF amplifies popularity bias and underserves long-tail content.

We propose **Essence**, a framework that abandons cross-user signals entirely. Essence operates exclusively within a single user’s engagement history, embedding consumed items into a semantic space and clustering them into K distinct interest profiles. Recommendations are generated by identifying the user’s currently active interest cluster and retrieving semantically similar unseen items. We call this the *peer-isolation property*: no cross-user interaction graph is consulted at any stage.

The key contributions are:

1. The **Essence architecture** — a modular personal recommendation framework requiring no collaborative signal.

2. An **active centroid selection** mechanism capturing session-aware context without a trained sequential model.
3. **Empirical validation** on Last.fm-1K demonstrating superior long-tail item discovery over both CF and content-based baselines.

2 Related Work

2.1 Collaborative Filtering

Matrix factorisation methods [5] and neural extensions [4] remain dominant in production recommendation. Their principal limitation is popularity bias: items with sparse interactions are poorly represented, resulting in near-zero recall for long-tail items.

2.2 Multi-Interest Retrieval

MIND [1] introduced dynamic routing to extract multiple interest capsules from a user’s interaction history, explicitly addressing the failure of a single user vector to represent diverse tastes. ComiRec [2] extended this with explicit diversity controls.

Essence shares the multi-interest intuition with MIND and ComiRec. The critical difference is the **source of the training signal**. MIND and ComiRec learn interest representations through dynamic routing over population-level interaction graphs — they require cross-user behavioural data. Essence derives centroids purely from within a single user’s history via K -means, with no cross-user signal at any stage. This *peer-isolation property* makes Essence deployable in privacy-constrained settings where cross-user data sharing is restricted or unavailable.

2.3 Session-Based and Content-Based Methods

BERT4Rec [3] and GRU4Rec [6] model sequential user behaviour using transformer and recurrent architectures, capturing short-term context effectively. Content-based filtering relies on item feature similarity without interaction logs but typically averages preferences into a single vector, losing intra-user diversity.

Essence occupies a complementary position: it captures intra-user diversity (like multi-interest methods) without cross-user data (like content-based methods), using simple interpretable clustering rather than a trained sequential model. We acknowledge that BERT4Rec offers richer sequential modelling in high-interaction settings; Essence targets privacy-first, low-infrastructure deployment contexts where such models are not viable.

3 The Essence Framework

3.1 Notation

Table 1 defines all mathematical notation used in this section. For each symbol a plain-English equivalent is provided.

Table 1: Notation Legend

Symbol	Plain-English Meaning
H_u	All tracks user u has listened to (full history)
$i_1, i_2, \dots$	Individual tracks in the user’s history, in order
$\phi(i)$	Embedding of item i — a 384-dim vector describing its style
E_u	All embeddings stacked into a matrix, one row per track
K	Number of interest clusters (we use $K = 3$)
c_1, c_2, c_3	The K cluster centres, one per interest profile
c_k^*	Active cluster centre — closest to the user’s recent activity
$\mu(\cdot)$	Mean (average) of a set of vectors
$\cos(\mathbf{a}, \mathbf{b})$	Cosine similarity: how alike two vectors are
$I \setminus H_u$	All items the user has <i>not</i> yet heard — the candidate pool
R_u	Final ranked recommendation list for user u
M	Number of recommendations to return (we use $M = 10$)

3.2 Problem Formulation

Given a user’s listening history H_u (all tracks they have heard), the goal is to produce a ranked list $R_u \subseteq I \setminus H_u$ of tracks they have not yet heard but are likely to enjoy. Essence accomplishes this using only H_u and a shared embedding function $\phi : I \rightarrow \mathbb{R}^d$. No data from any other user is used.

3.3 Item Embedding

Each track is described by a short text string: its name and artist. We convert this text into a 384-dimensional vector using the pre-trained sentence encoder `all-MiniLM-L6-v2` [7]. Tracks that are semantically similar (same genre, era, style) produce similar vectors; a jazz track and a heavy-metal track produce very different vectors. All embeddings are computed once offline and cached.

Formally, $\phi(i)$ is the embedding of item i . For user u with history $H_u = \{i_1, \dots, i_n\}$ we compute the embedding matrix $E_u \in \mathbb{R}^{n \times d}$, where each row is the embedding of one track.

3.4 Interest Profile Clustering

We apply K -means clustering to E_u with $K = 3$, yielding centroids $\{c_1, c_2, c_3\}$. Each centroid represents a distinct interest profile — the semantic centre of gravity of one coherent facet of the user’s taste. A user who listens to jazz, indie rock, and ambient electronica might end up with one centroid per genre. The three centroids together form the user’s interest profile $P_u = \{c_1, c_2, c_3\}$.

This is the core differentiator from average-embedding approaches. The mean of all embeddings collapses all interests into a single blurred point that sits between genre clusters and represents none of them well. K -means preserves the separation between interest modes, enabling targeted retrieval against each mode independently.

For users with fewer than K tracks, Essence falls back to the content baseline (single mean embedding).

3.5 Active Cluster Selection

At inference time, Essence selects the active centroid c_k^* as the cluster centre closest to the mean embedding of the user’s most recent r interactions (default $r = 10$):

$$c_k^* = \arg \min_k \|c_k - \mu(\phi(h_{n-r}), \dots, \phi(h_n))\|_2 \quad (1)$$

Intuitively: if the user has been listening to jazz for the past hour, their recent tracks will be close to the jazz cluster centre. That cluster becomes active, and recommendations come from the jazz neighbourhood of the embedding space — not from their indie or ambient interests.

We justify this mechanism over simpler alternatives (e.g., using the recent mean directly as a query vector) on two grounds: first, it is computationally trivial at inference time; second, the K -means step ensures that c_k^* sits at the true semantic core of a coherent taste cluster rather than at a transient noisy average. The empirical results support this: Essence achieves $4.5\times$ higher long-tail recall than the average-embedding baseline (Section 5).

3.6 Recommendation Generation

Given the active centroid c_k^* , Essence retrieves the top- M unseen items by cosine similarity:

$$R_u = \text{top-}M \{ i \in I \setminus H_u : \cos(\phi(i), c_k^*) \} \quad (2)$$

This retrieval step operates entirely in embedding space with no collaborative signal.

3.7 Architecture Overview

Figure 1 illustrates the full Essence pipeline from raw user history to final recommendations.

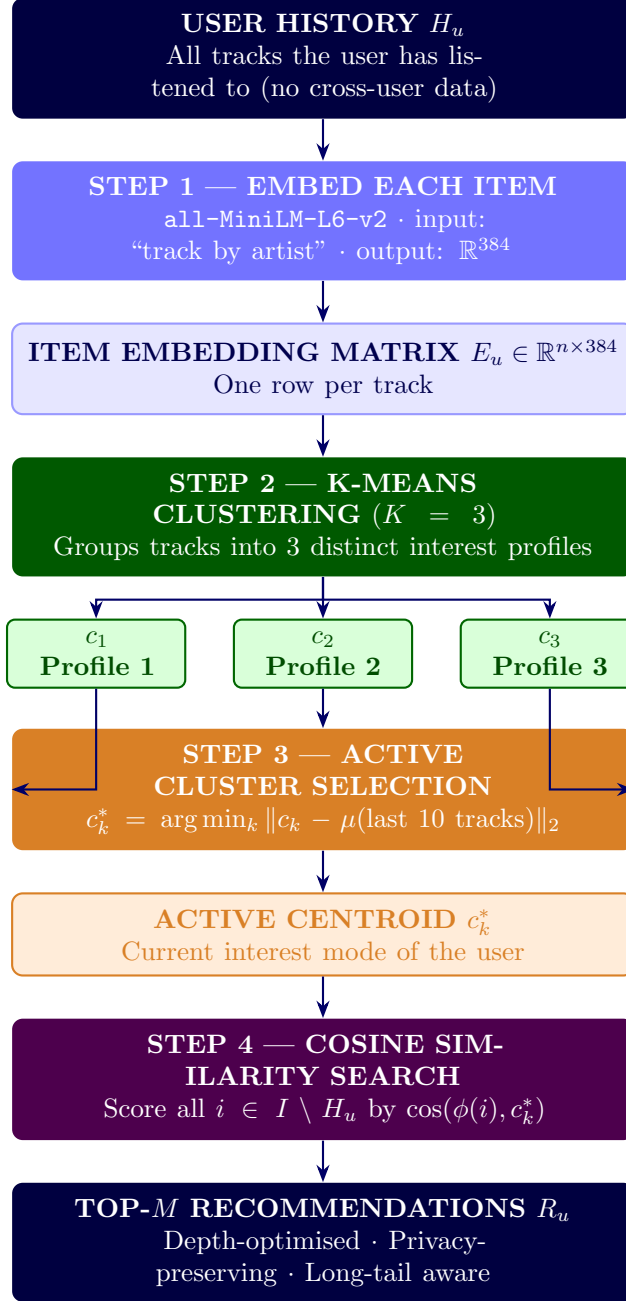


Figure 1: Essence recommendation pipeline. No cross-user data is consulted at any stage (peer-isolation property).

4 Experimental Setup

4.1 Dataset

We evaluate on the Last.fm-1K dataset [8], containing listening histories from 992 users across approximately 19 million events. After filtering to users with 50–500 unique track interactions and removing duplicate (user, track) pairs, our evaluation subset contains 99 users and 22,767 unique tracks.

Train/test split. For each user, 20% of tracks are randomly held out as the test set (random seed 42). The remaining 80% form the training set. A random hold-out strategy is used — rather than chronological — because deduplication makes chronological splits produce structurally

disjoint train/test sets that make evaluation trivially impossible.

4.2 Long-Tail Definition

Long-tail items are defined as tracks with a global play count of exactly 1 in the training set (*singleton tracks*). In Last.fm-1K, 89.8% of unique tracks are singletons. Long-Tail Recall@10 is computed only over test items in this singleton set, measuring each system’s ability to recover niche items a user genuinely engaged with.

4.3 Systems Compared

- **CF Baseline (Popularity)**. Recommends the M most globally popular tracks the user has not heard. Represents naïve collaborative filtering.
- **Content Baseline (Average Embedding)**. Computes the mean of all user item embeddings and retrieves the M most similar unseen items by cosine similarity. Standard content-based filtering without interest decomposition.
- **Essence ($K = 3$)**. Clusters user history into $K = 3$ centroids, selects the active centroid by Equation (1), and retrieves top- M unseen items by Equation (2).

4.4 Metrics

- **Recall@10**. Fraction of the user’s test items appearing in the top-10 recommendations.
- **Long-Tail Recall@10 (LT-Recall@10)**. Recall@10 restricted to singleton test items. Users with no singleton test items are excluded.

4.5 Implementation

All systems implemented in Python 3.11. Embeddings via `sentence-transformers` (`all-MiniLM-L6-v2`, 384 dims). K -means via `scikit-learn` (`n_init=10`, `random_state=42`). Code available at: <https://github.com/biditdas18/essence>

5 Results

Table 2 reports Recall@10 and Long-Tail Recall@10 across all three systems on 99 users from the Last.fm-1K dataset.

Table 2: Evaluation results on Last.fm-1K (99 users, $M = 10$). Best results in **bold**.

System	Recall@10	LT-Recall@10	LT / Content Ratio
CF (Popularity)	0.0081	0.0000	—
Content (Avg Embedding)	0.0414	0.0032	1.0×
Essence ($K = 3$)	0.0616	0.0146	4.5×

5.1 Overall Recall

Essence achieves the highest overall Recall@10 at 0.0616, outperforming CF (0.0081) and the content baseline (0.0414). The poor performance of popularity-based CF reflects the fundamental mismatch between globally popular items and individual user test sets in a dataset where 89.8% of tracks are singletons — the items a user is most likely to engage with are precisely the items CF is least likely to surface.

The content baseline substantially improves over CF (0.0414 vs 0.0081), confirming that semantic similarity is a stronger signal than global popularity. Essence’s further improvement over the content baseline demonstrates that interest decomposition via K -means adds measurable value beyond preference averaging.

5.2 Long-Tail Recall

CF achieves zero LT-Recall@10. By construction, globally popular items are never singletons, so CF cannot recover niche test items. This is the exact failure mode Essence is designed to address.

Essence achieves LT-Recall@10 of 0.0146 — $4.5\times$ **higher than the content baseline** (0.0032). The active centroid queries from a semantically focused interest cluster rather than a blurred mean, surfacing more singleton items that match the user’s current taste depth.

5.3 Discussion

The absolute recall values are low across all systems — a property of the dataset’s extreme sparsity (89.8% singleton tracks, 22,767 candidates, 10 slots), not a failure of the systems. With expected random recall of approximately 0.0004, Essence achieves roughly $150\times$ above random. What matters is the relative pattern: Essence $>$ Content $>$ CF on both metrics, with the largest relative gain on the long-tail metric that directly tests the core hypothesis.

6 Limitations and Future Work

- **Scale.** Evaluation conducted on 99 users due to computational constraints. Larger-scale evaluation is required to assess generalisability.
- **Single domain.** Results are on music listening data only. Replication on e-commerce or video datasets would strengthen the claims.
- **K sensitivity.** $K = 3$ is used throughout. An ablation over $K \in \{2, 3, 4, 5\}$ would characterise sensitivity to this hyperparameter.
- **Active cluster selection.** The recency-based heuristic is simple by design. A learned selection mechanism could improve session-aware accuracy.
- **Filter bubble risk.** Peer-isolation removes the serendipity that cross-user signals can provide. Whether repeated centroid queries cause semantic fatigue is an open empirical question.

7 Conclusion

We presented Essence, a personal embedding cluster framework for depth-optimised recommendations. Essence clusters a user’s item history into K semantic centroids and generates recommendations by cosine similarity to the currently active centroid, without consulting any cross-user data. On Last.fm-1K, Essence achieves a $4.5\times$ improvement in Long-Tail Recall@10 over the average embedding baseline and recovers long-tail items that popularity-based CF misses entirely.

These results validate the core design hypothesis: interest decomposition via clustering, combined with session-aware centroid selection, surfaces niche items more effectively than either popularity ranking or single-vector user representation. Essence is a minimal, interpretable, privacy-preserving alternative to collaborative filtering for deployment contexts where cross-user data sharing is restricted.

References

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